

Introduction to Mathematics and Optimisation

Exercises Week 8

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1 Exercise 6.50

Show that $x_0 = 0$ is a critical point of the function $f : (-\frac{1}{2}, \infty) \rightarrow \mathbb{R}$ defined by

$$f(x) = e^x + \ln(1 + 2x) - 3x.$$

Use Theorem 6.47 in deciding if it is local maximum, minimum or neither.

1.1 Solution

Critical point x_0 is at $f'(x_0) = 0$. We first need to find the derivative of the function f . Using the rules from section 6.3.4 we arrive at $f'(x) = e^x + \frac{2}{1+2x} - 3$ if we evaluate this at $f'(0)$ we get the value $e^0 + \frac{2}{1} - 3 = 0$ which means $x_0 = 0$ is a critical point.

By Theorem 6.47 we need to check higher order derivatives until we find one where the $f^{(n)}(x_0) \neq 0$. We start by finding the second order derivative which is equal to $f''(x) = e^x - \frac{4}{(1+2x)^2}$. If we evaluate it at x_0 we get the value $e^0 - \frac{4}{1^2} = -3$. Since $f''(x_0) \neq 0 \wedge f''(x_0) < 0$ and it's even derivative the critical point x_0 is a local maximum.

2 Exercise 6.57

Show that $f : \{x \in \mathbb{R} | x \geq 0\} \rightarrow \mathbb{R}$ given by

$$f(x) = -\sqrt{x}$$

is a strictly convex function.

2.1 Solution

Corollary 6.52 states that $f : (a, b) \rightarrow \mathbb{R}$ is strictly convex if $\forall x \in (a, b) : f''(x) > 0$. First derivative of f is $f'(x) = -\frac{1}{2}x^{-\frac{1}{2}} = -\frac{1}{2\sqrt{x}}$ and second derivative is equal to $f''(x) = \frac{1}{4}x^{-\frac{3}{2}} =$

$\frac{1}{4x\sqrt{x}}$. Since we are operating on the interval $(0, \infty)$ x is always positive and so is $\sqrt{x}$ therefore so is the whole expression. Therefore by Corollary 6.52 the function f is strictly convex on the interval $(0, \infty)$.

To satisfy the Definition 6.1 of strict convexity we need to show that $f((1-t)u + tv) < (1-t)f(u) + tf(v)$ holds for $t \in (0, 1)$ and $u, v \in [0, \infty)$ where $u \neq v$. We already know by Corollary 6.52 that it holds for any two points in the interval $(0, \infty)$ so we only need to show that it holds for $u = 0$ and $v \in (0, \infty)$.

$$\begin{aligned}
f((1-t)u + tv) &< (1-t)f(u) + tf(v) \\
f((1-t)0 + tv) &< (1-t)f(0) + tf(0) \\
f(tv) &< (1-t)0 + tf(v) \\
f(tv) &< tf(v) \\
-\sqrt{tv} &< -t\sqrt{v} \\
-\sqrt{t}\sqrt{v} &< -t\sqrt{v} \\
-\sqrt{t} &< -t \text{ as } \sqrt{v} \text{ is strictly positive on interval } (0, \infty) \\
-\sqrt{t} &< -(\sqrt{t})^2 \\
1 &> \sqrt{t}
\end{aligned}$$

which is always true for t in interval $(0, 1)$. Therefore the function f is strictly convex on the interval $[0, \infty)$

3 Exercise 6.59

Suppose that $f : (a, b) \rightarrow \mathbb{R}$ is a differentiable convex function and $x_0 \in (a, b)$ is a critical point for f . What can you say about x_0 using Theorem 6.58?

3.1 Solution

Since $f'(x_0) = 0$ we can substitute x_0 into Theorem 6.58 receiving the inequity

$$\begin{aligned}
f(y) &\geq f(x_0) + f'(x_0)(y - x_0) \\
f(y) &\geq f(x_0)
\end{aligned}$$

This is the same inequity in Definition 6.9 defining global minimum. So x_0 is a global minimum of function f on interval (a, b)